

DEPLOYMENT HANDBOOK

Real-World Evidence / HEOR Deploying on AWS

Step-by-step, console and CLI — from an empty AWS account to a running, governed, human-gated agent. Tailored edition: AgentId **07-rwe-heor**; approval gate **EPIDEMIOLOGIST**.

Systems-Integrator field & delivery reference · June 2026

Illustrative reference — figures are mockups, not actual AWS screenshots. Validate for your environment.

Tailored edition for Real-World Evidence / HEOR (AgentId 07-rwe-heor). This is the companion to the suite master handbook (docs/DEPLOYMENT-HANDBOOK.md); it carries this agent's systems, approver role, workflow, and console figures. The phases and controls are identical across all eight agents.

0. Two deployment shapes

Shape	What runs	When to choose
Container lift	The LangGraph agent image on Bedrock AgentCore Runtime (or ECS Fargate)	Fastest; agent code unchanged
Native rebuild (this book)	Deterministic steps in Lambda , drafting via Strands/Bedrock , orchestrated by Step Functions with a <code>waitForTaskToken</code> human gate	Highest fidelity to managed-serverless; clearest audit & HITL story

Both deploy through the same CloudFormation quickstart with `AgentId=07-rwe-heor` , and both put every system-of-record call behind the **MCP authorization gateway = Bedrock AgentCore Gateway + Identity**.

This agent at a glance — systems: this edition connects RWD and related systems; the **approval gate** is the **EPIDEMIOLOGIST**; high-risk tools requiring human approval: none (reads / reversible actions; the finalize step is still human-approved). Time budget: ~2–4 hours for a first dev deployment once prerequisites are granted.

1. Prerequisites

You need	Who provides it	Notes
AWS account/sandbox with admin (or the scoped deploy policy)	Customer cloud team	Start in dev, never prod
Amazon Bedrock model access (Claude)	Account admin	One-time per account+region (Phase 3.1)
An AgentCore-enabled Region	Architect	e.g. <code>us-east-1</code>
Enterprise IdP metadata (Okta/Entra/AD)	Customer identity team	SAML/OIDC metadata URL
The RWD endpoint + an API token	Customer system owner	Deploy against the bundled reference service first, then swap
Local tooling	You	AWS CLI v2, Docker/Finch (ARM64), Python 3.11, <code>zip</code>

Confirm before you touch the console: who is the named **EPIDEMIOLOGIST** for this agent's human gate, and who signs computer-system validation at go-live.

2. Pre-flight (local)

```
aws sts get-caller-identity           # right account?
cd 07-rwe-heor-agent && python -m venv venv && . venv/bin/activate
pip install -r requirements.txt && pip install -e ../platform_core
EXTRACT_MODE=demo pytest tests/ -q    # the agent is healthy offline first
```

3. Phase 1 — Account foundation

3.1 Enable Bedrock model access (Console)

Bedrock → **Model access** → **Manage model access** → check the Anthropic Claude tiers → **Save**; wait for **Access granted**. Confirm the Region (top-right) is AgentCore-enabled.

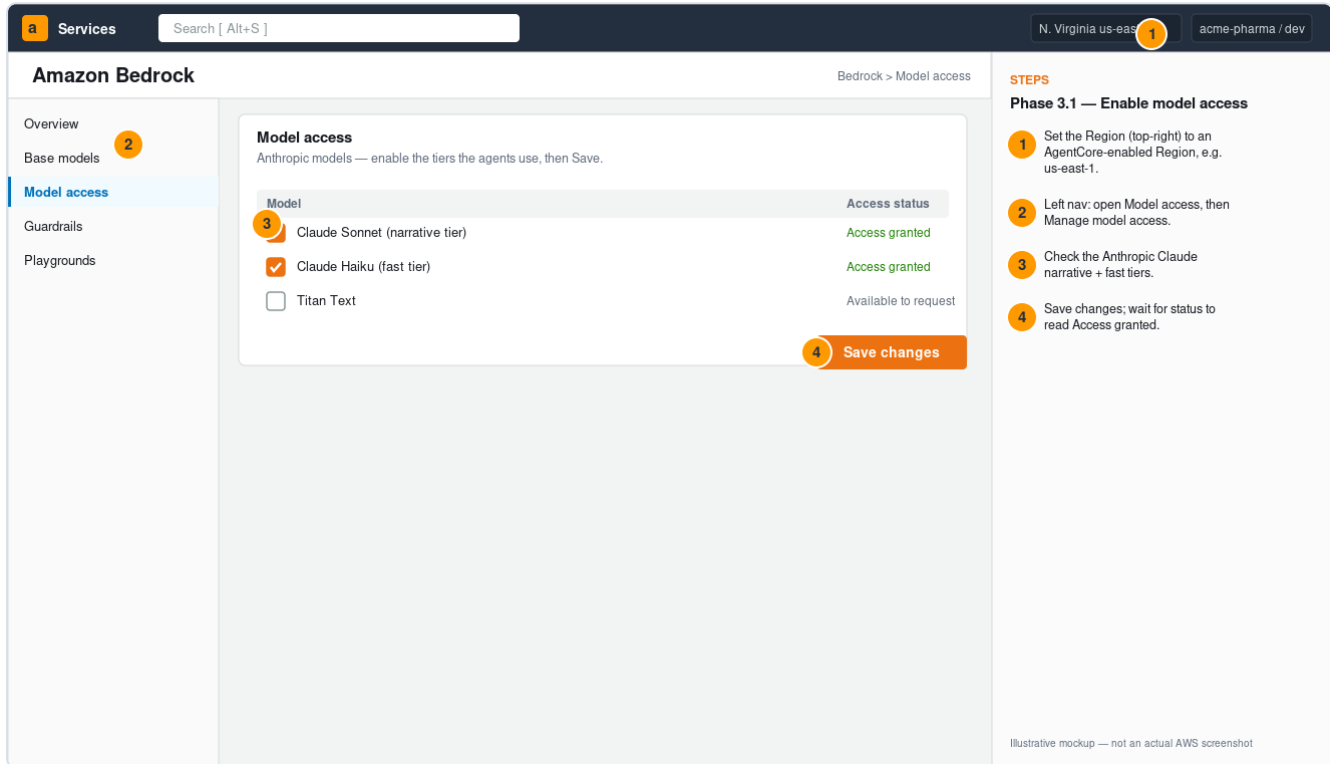


Figure 1. Amazon Bedrock → Model access (illustrative mockup).

3.2 Create a Bedrock Guardrail (the CloudFormation quickstart creates this)

PHI filters (SSN→Block; Email/Name/Phone→Anonymize) + a denied topic for off-label/absolute claims.
Record the **Guardrail ID**.

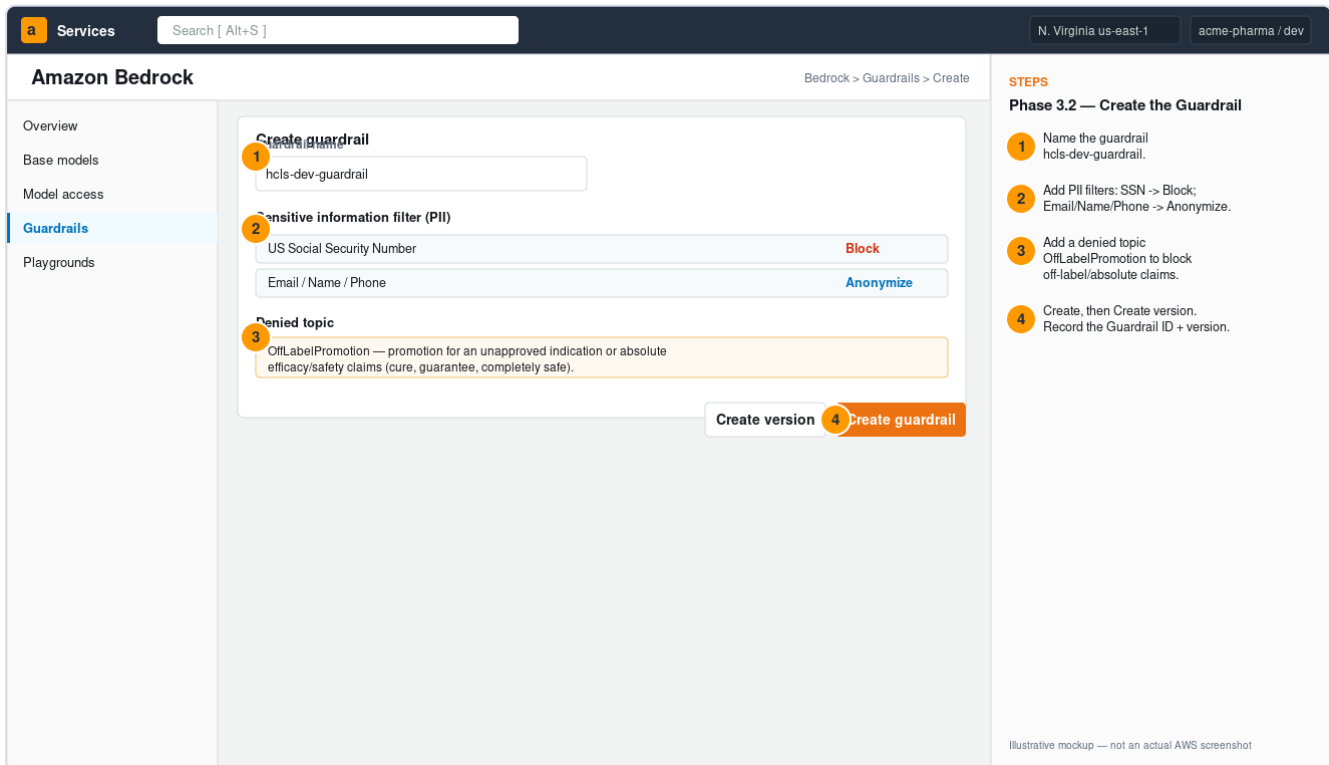


Figure 2. Bedrock → Guardrails (illustrative mockup).

3.3 Cognito + IdP federation — map the approver

Federate the customer IdP and map the approver group to `custom:hcls_role`. For this agent, map `GRP-RWE-EPIDEMIOLOGISTS` → `EPIDEMIOLOGIST`.

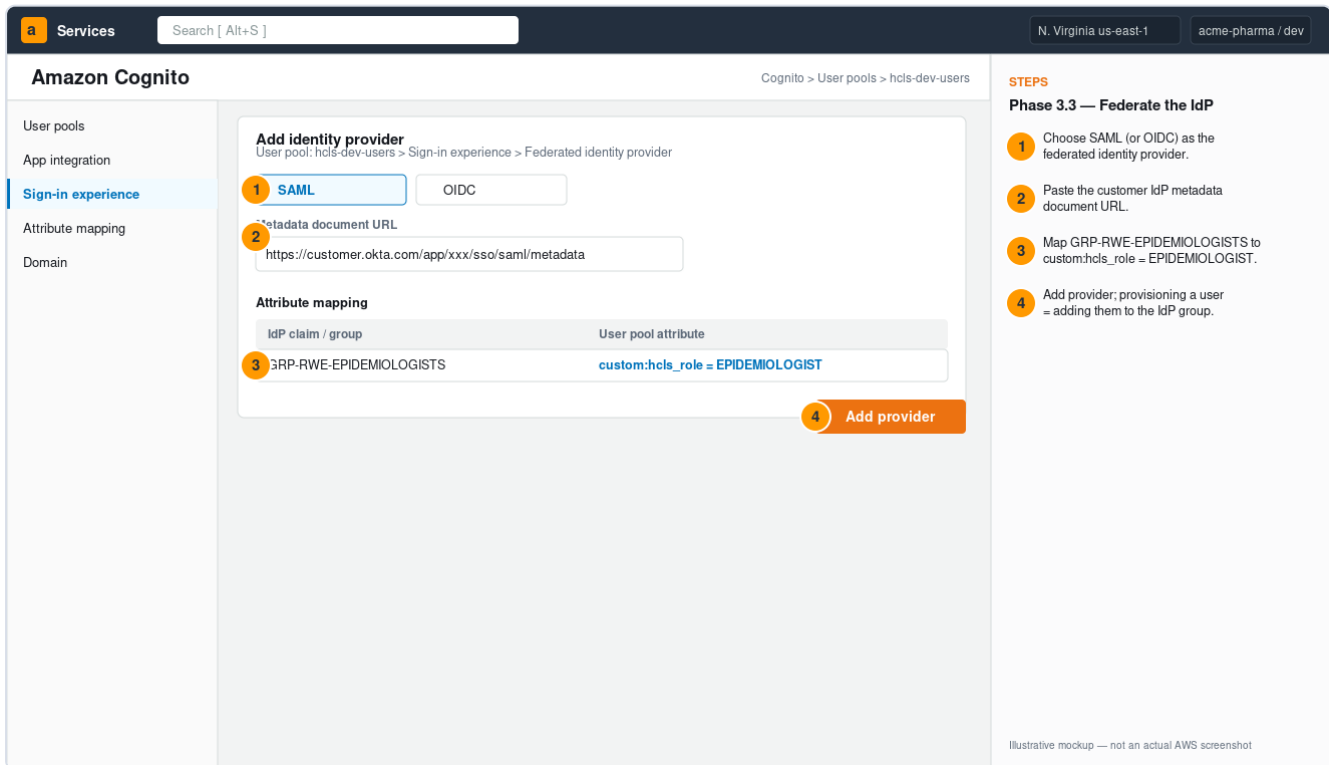


Figure 3. Cognito → Sign-in experience: map GRP-RWE-EPIDEMOLOGISTS → EPIDEMOLOGIST (illustrative mockup).

3.4 KMS key (quickstart creates `alias/hcls-dev`) — per-environment CMK encrypting the audit table, review table, and WORM bucket.

4. Phase 2 — Stage artifacts

```
aws s3 mb s3://my-cfn-bucket
aws s3 cp infra/cloudformation/ s3://my-cfn-bucket/hcls/ --recursive --exclude "*" --include "*.yaml"
cd aws-native-reference/07-rwe-heor && mkdir -p build
zip -r build/lambda.zip core.py strands_agent.py lambda/ requirements.txt
aws s3 mb s3://my-code-bucket
aws s3 cp build/lambda.zip s3://my-code-bucket/07-rwe-heor/lambda.zip
```

(Container path: build the ARM64 image from `aws-native-reference/_shared/runtime` and push to ECR.)

5. Phase 3 — Deploy the infrastructure (CloudFormation)

CloudFormation → **Create stack** → template S3 URL → set parameters (**AgentId** 07-rwe-heor , Environment, DeployMode, code bucket, IdP URL) → check the IAM capability box → **Submit** → watch Events to **CREATE_COMPLETE** → read **Outputs** (GatewayId , GuardrailId , AuditTable).

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AWS CloudFormation CloudFormation > Create stack

Stacks
Create stack
StackSets
Exports
Designer

Create stack

1 Amazon S3 template URL
https://my-cfn-bucket.s3.amazonaws.com/hcls/quickstart.yaml

Stack name
hcls-dev-07-rwe-heor

2 **Parameters**

Environment	dev
AgentId	07-rwe-heor
DeployMode	native
LambdaCodeBucket	my-code-bucket
IdpMetadataUrl	https://customer.okta.com/.../metadata

3 ☒ I acknowledge that AWS CloudFormation might create IAM resources with custom names.

4 **Submit**

STEPS
Phase 5 — Deploy with CloudFormation

- 1 Specify template: the quickstart.yaml S3 URL.
- 2 Set AgentId = 07-rwe-heor (plus Environment, DeployMode, code bucket, IdP URL).
- 3 Check the IAM capability acknowledgement.
- 4 Submit; watch Events to CREATE_COMPLETE; read Outputs.

Illustrative mockup — not an actual AWS screenshot

Figure 4. CloudFormation → Create stack for AgentId 07-rwe-heor (illustrative mockup).

```
aws cloudformation deploy --template-file infra/cloudformation/quickstart.yaml \
  --stack-name hcls-dev-07-rwe-heor --capabilities CAPABILITY_NAMED_IAM \
  --parameter-overrides Environment=dev AgentId=07-rwe-heor DeployMode=native \
  TemplateBaseUrl=https://my-cfn-bucket.s3.amazonaws.com/hcls \
  LambdaCodeBucket=my-code-bucket LambdaCodeKey=07-rwe-heor/lambda.zip \
  IdpMetadataUrl=https://customer.okta.com/app/xxx/sso/saml/metadata
```

Set Environment=prod in production: the Guardrail becomes mandatory (the LLM factory refuses to start un-guardrailed).

6. Phase 4 — The MCP layer (AgentCore Gateway + Identity)

The `agentcore-gateway.yaml` nested stack created the Gateway (JWT-authorized by your Cognito pool) and one **target per system of record** this agent uses. High-risk targets (none (reads / reversible actions; the finalize step is still human-approved)) require a human-approval token — enforced by the `waitForTaskToken` gate in Phase 8. Deny-by-default authorization (agent grant $\cap$ user entitlement), scoped tokens, and PHI-masked audit match the Python reference in `platform_core`.

7. Phase 5 — Configure (connect RWD)

7.1 Store the credential (Secrets Manager)

Store the RWD API token as `hcls/rwd_api_token` (key `rwd_api_token`). The `hcls/` prefix is what `get_secret()` resolves; the agent role has `GetSecretValue` on `hcls/*`.

STEPS

Phase 7 — Store the system credential

- 1 Choose Other type of secret.
- 2 Add key `rwd_api_token` = the RWD API token.
- 3 Name the secret `hcls/rwd_api_token` (the `hcls/` prefix matters).
- 4 Store. The agent role has `GetSecretValue` on `hcls/*`.

Illustrative mockup — not an actual AWS screenshot

Figure 5. Secrets Manager → Store `hcls/rwd_api_token` (illustrative mockup).

7.2 Point at live inference + live systems

Variable	Value	Effect
<code>LLM_PROVIDER</code>	<code>bedrock</code>	In-account inference (no data egress)
<code>BEDROCK_GUARDRAIL_ID</code>	from stack Outputs	PHI + off-label filter
<code>CONNECTOR_MODE</code>	<code>live</code>	Real connectors, not fixtures
<code>RWD_BASE_URL</code>	the RWD base URL	Where the live connector calls
<code>ENVIRONMENT</code>	<code>prod</code> (in prod)	Makes the Guardrail mandatory

This agent's systems / connectors:

kind	System	Base URL env
rwd	Claims / registry / RWD platform	RWD_BASE_URL

Tip: deploy first with `RWD_BASE_URL` pointed at a reference/staging endpoint to validate the pipeline, then swap to the customer's production RWD — a one-variable change, no code change.

8. Phase 6 — Smoke test + the human gate

Step Functions → open `hcls-dev-07-rwe-heor` → **Start execution** with the agent's `sample_input.json`. The graph runs **Assemble** → **Synthesize** → **Check** → **RouteChoice** → **HumanReviewGate** → **Finalize** and **pauses at HumanReviewGate** — the `waitForTaskToken` gate for the **EPIDEMIOLOGIST**. It is supposed to wait; nothing finalizes without a human.

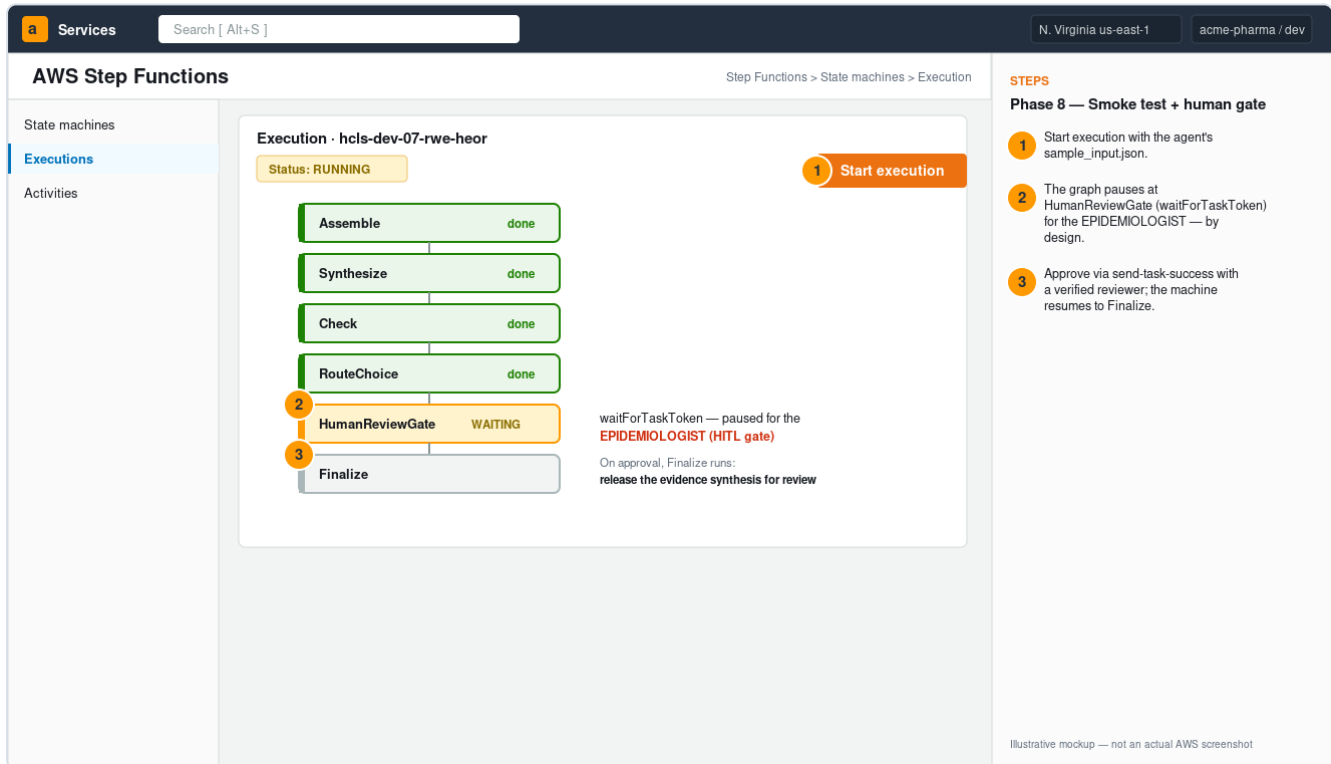


Figure 6. Execution paused at **HumanReviewGate** for the **EPIDEMIOLOGIST** (illustrative mockup).

```
aws stepfunctions send-task-success --task-token "<token-from-review-table>" \
  --task-output '{"approved": true, "reviewer": {"sub": "approver-1",
"custom:hcls_role": "EPIDEMIOLOGIST"}}'
```

On approval the machine resumes to **Finalize**, which will release the evidence synthesis for review (gateway-authorized) and seal the audit trail. To reject, use `send-task-failure`.

8.3 Confirm the audit trail (DynamoDB)

Open `hcls-dev-audit` → **Explore items**: append-only, PHI-masked entries per step, with the **EPIDEMIOLOGIST** bound to the approved action. This is your 21 CFR Part 11 evidence.

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DynamoDB > Tables > hcls-dev-audit > Explore items

Tables

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Explore items · hcls-dev-audit

ts	decision	tool	user	args (PHI-masked)
13:20:05	ALLOW	nwd.run_cohort_query	u-heor-1	cohort defn (de-identified)
13:20:40	ALLOW	nwd.run_cohort_query	u-heor-1	data-quality pass
13:41:12	ALLOW	finalize.release_synthesis	u-epi-2	approved_by=u-epi-2

Append-only by IAM policy (UpdateItem/DeleteItem denied) — tamper-evident, 21 CFR Part 11.

STEPS

Phase 8.3 — Confirm the audit trail

- Each workflow step writes an append-only, PHI-masked audit item.
- Reads and reversible actions are logged with full lineage.
- The human-approved finalize is ALLOW with the reviewer bound.

Illustrative mockup — not an actual AWS screenshot

Figure 7. Append-only, PHI-masked audit trail (illustrative mockup).

9. Go-live checklist

- [] Bedrock model access granted; Guardrail attached and tested (PHI string blocked/anonymized).
- [] IdP federation live; `GRP-RWE-EPIDEMIOLOGISTS` maps to `EPIDEMIOLOGIST`; a non-approver is correctly denied the high-risk action.
- [] `CONNECTOR_MODE=live` against RWD; a read and a (test) write both appear in the audit trail.
- [] HITL gate verified: execution pauses; finalize only after `send-task-success` with a verified EPIDEMIOLOGIST.
- [] Audit trail append-only (update/delete denied); WORM retention set.
- [] `pytest` + governance evals captured as validation evidence; prompt manifest matches.
- [] Customer CSV/CSA signed for the intended use; penetration test scheduled/done.
- [] Runbooks reviewed: incident, DR, HITL-queue, model-degradation.

10. Troubleshooting

Symptom	Likely cause	Fix
<code>AccessDeniedException</code> on Bedrock	Model access not granted in Region	Phase 3.1
Stack fails on <code>AWS::BedrockAgentCore::*</code>	Region lacks AgentCore	Use an AgentCore Region, or the API Gateway + Lambda-authorizer + Cognito fallback
LLM factory refuses to start	<code>ENVIRONMENT=prod</code> but no <code>BEDROCK_GUARDRAIL_ID</code>	Set the Guardrail ID (intentional guard)
Execution stuck at <code>HumanReviewGate</code>	Working as designed	Send <code>send-task-success</code> with a verified EPIDEMIOLOGIST
Connector <code>NotImplementedError</code>	<code>CONNECTOR_MODE=live</code> but that system's live connector isn't wired	Implement typed methods in <code>connectors/live.py</code> (mirror <code>LiveSafetyConnector</code>)
Tool call denied unexpectedly	User role lacks entitlement / agent not granted tool	Check <code>ROLE_ENTITLEMENTS</code> / <code>AGENT_TOOL_GRANTS</code> + Cognito mapping

Companion to the suite master handbook ([docs/DEPLOYMENT-HANDBOOK.md](#)), [SOLUTION-FIELD-GUIDE.md](#) (sales/SA), and this agent's [docs/](#) set. For operations after go-live, see [runbooks/](#).